

Uputstvo za A5 Difoltne rute:

1. Zadatak je pripremljen sa javnim IP adresama, a navedeno je da se za novi LAN u Beogradu koriste privatne IP adrese. Mreža je organizovana tako da se glavni server nalazi u Beogradu, u istom delu mreže u odnosu na FrameRelay oblak. DNS server preduzeća 157.104.2.10 povezan je preko R2. Zato počinjemo sa analizom IP adresa u mreži.

Ruter Beograd:

```
Beograd#sh ip interf brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          157.104.1.100   YES manual up          up
Ethernet1/0               157.104.200.100 YES manual up          up
Ethernet1/1               152.104.200.100 YES manual up          down
Ethernet1/2               unassigned      YES unset  administratively down down
Ethernet1/3               unassigned      YES unset  administratively down down
Beograd#
```

Interfejs ka R1 – spoljnjoj mreži	Interfejs ka ServerPT – www.prevoz.rs	Interfejs – slobodan (javnaIP)
<pre>interface FastEthernet0/0 ip address 157.104.1.100 255.255.255.0 router rip network 157.104.0.0</pre>	<pre>interface Ethernet1/0 ip address 157.104.200.100 255.255.255.0</pre>	<pre>interface Ethernet1/1 ip address 152.104.200.100 255.255.255.0</pre>

2. Treba u Beogradu dodati novi LAN sa privatnim adresama i automatskom dodelom adresa.

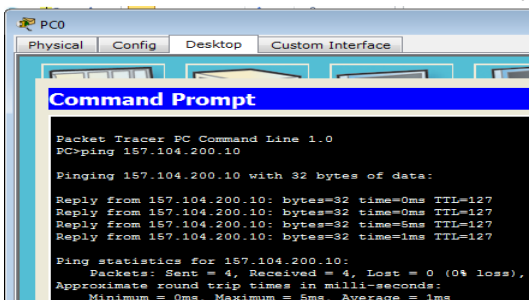
```
interface Ethernet1/2
ip address 192.168.200.1 255.255.255.0
no shutdown
```

```
ip dhcp pool LAN
network 192.168.200.0 255.255.255.0
default-router 192.168.200.1
dns-server 157.104.2.10
ip dhcp excluded-address 192.168.200.1
```

3. Komandom: **show running-conf** proverimo da smo dobro konfigurisali R1.

<pre>ip dhcp excluded-address 192.168.200.1 ! ip dhcp pool LAN network 192.168.200.0 255.255.255.0 default-router 192.168.200.1 dns-server 157.104.2.10</pre>	<pre>interface Ethernet1/2 ip address 192.168.200.1 255.255.255.0</pre>
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4. U lokalni LAN dodamo računar PC0, podesimo da dobije IP adresu preko DHCP-a, proverimo konfiguraciju i pokušamo PING ka serveru 157.104.200.10 (www.prevoz.rs). Treba da je uspešno pošto se radi o mrežama istog rutera Beograd.



5. Ako pokušamo PING između PC0 (192.168.200.2) i rutera R1, vidimo da je neuspešno, pošto R1 nema tu mrežu u svojoj tabeli ruta.

```
R1>ena
R1#sh ip route
157.104.0.0/24 is subnetted, 5 subnets
C    157.104.1.0 is directly connected, FastEthernet0/0
R    157.104.2.0 [120/1] via 157.104.100.2, 00:00:11, Serial0/0.1
R    157.104.3.0 [120/1] via 157.104.100.3, 00:00:20, Serial0/0.1
C    157.104.100.0 is directly connected, Serial0/0.1
R    157.104.200.0 [120/1] via 157.104.1.100, 00:00:06, FastEthernet0/0
```

6. Treba dodati podrazumevanu rutu ka ruteru u Beogradu:

```
ip route 0.0.0.0 0.0.0.0 157.104.1.100
```

```
R1#sh ip route
```

```
Gateway of last resort is 157.104.1.100 to network 0.0.0.0
```

```
157.104.0.0/24 is subnetted, 5 subnets
C    157.104.1.0 is directly connected, FastEthernet0/0
R    157.104.2.0 [120/1] via 157.104.100.2, 00:00:11, Serial0/0.1
R    157.104.3.0 [120/1] via 157.104.100.3, 00:00:16, Serial0/0.1
C    157.104.100.0 is directly connected, Serial0/0.1
R    157.104.200.0 [120/1] via 157.104.1.100, 00:00:20, FastEthernet0/0
* 0.0.0.0 [120/1] via 157.104.1.100
```

Proveriti PING od 192.168.200.2 do 157.104.1.1.

Sada treba da je uspešan!

7. Proveriti vezu ka DNS serveru preduzeća na adresi 157.104.2.10 komandom **tracert**
Pošto nema rute ka 192.168.200.0/24 na ruteru R2 dobijamo sledeće:

```
PC>tracert 157.104.2.10

Tracing route to 157.104.2.10 over a maximum of 30 hops:

  1  0 ms    1 ms    1 ms    192.168.200.1
  2  2 ms    1 ms    0 ms    157.104.1.1
  3  *        *        *        Request timed out.
  4  *        *        *        Request timed out.
  5  *        *        *        Request timed out.
```

R2# ip route prikazuje sledeće rute:

```
Gateway of last resort is not set

157.104.0.0/24 is subnetted, 5 subnets
R    157.104.1.0 [120/1] via 157.104.100.1, 00:00:03, Serial0/0.2
C    157.104.2.0 is directly connected, FastEthernet0/0
R    157.104.3.0 [120/1] via 157.104.100.3, 00:00:10, Serial0/0.2
C    157.104.100.0 is directly connected, Serial0/0.2
R    157.104.200.0 [120/2] via 157.104.100.1, 00:00:03, Serial0/0.2
```

8. Na R1 treba konfigurisati u **Router rip**:
default-information originate

9. Sada tabela ruta na R2 izgleda ovako: A **tracert** sa PC na DNS server ovako:

```
Gateway of last resort is 157.104.100.1 to network 0.0.0.0

157.104.0.0/24 is subnetted, 5 subnets
R    157.104.1.0 [120/1] via 157.104.100.1, 00:00:05, Serial0/0.2
C    157.104.2.0 is directly connected, FastEthernet0/0
R    157.104.3.0 [120/1] via 157.104.100.3, 00:00:01, Serial0/0.2
C    157.104.100.0 is directly connected, Serial0/0.2
R    157.104.200.0 [120/2] via 157.104.100.1, 00:00:05, Serial0/0.2
R*  0.0.0.0/0 [120/1] via 157.104.100.1, 00:00:05, Serial0/0.2
```

```
PC>tracert 157.104.2.10

Tracing route to 157.104.2.10 over a maximum of 30 hops:

  1  0 ms    0 ms    1 ms    192.168.200.1
  2  1 ms    0 ms    0 ms    157.104.1.1
  3  2 ms    2 ms    0 ms    157.104.100.2
  4  0 ms    5 ms    0 ms    157.104.2.10
```

10. Default route pojavljuje se i na R3:

```
Gateway of last resort is 157.104.100.1 to network 0.0.0.0

157.104.0.0/24 is subnetted, 5 subnets
R    157.104.1.0 [120/1] via 157.104.100.1, 00:00:11, Serial0/0.3
R    157.104.2.0 [120/1] via 157.104.100.2, 00:00:10, Serial0/0.3
C    157.104.3.0 is directly connected, FastEthernet0/0
C    157.104.100.0 is directly connected, Serial0/0.3
R    157.104.200.0 [120/2] via 157.104.100.1, 00:00:11, Serial0/0.3
R*  0.0.0.0/0 [120/1] via 157.104.100.1, 00:00:11, Serial0/0.3

R3#
```

Treba proveriti vezu sa PC do WEB servera 157.104.3.10 (i /ili tabelu ruta)

11. Da bi svič bio dostupan iz spoljašnje mreže, *mora imati IP adresu iz privatne mreže npr. 192.168.200.0/24. dodeljenu loopback interfejsu. OVO NE MOŽE U CPT 6.2*

Dalje, moraju biti definisane lozinke konzolna, enable(secret) i vty.

Za Telnet:
line vty 0 4
password telnet
login

Za konzolu
line con 0
password console
login

Za administratorski – povlašćen pristup:
enable secret admin
enable password cisco

Za dodelu IP adrese:–
Interface– loopback 1
ip address– 192.168.200.3 255.255.255.0

12. Dodati bežični ruter  za laptop u Loznici. Povezati WiFi ruter na oblak

Povezati ruter preko DSL modema . U oblaku proveriti da je odgovarajući Modem povezan na pravi Eth

Signali za portove biće zeleni i ako promašimo vezu!

DSL

Modem4 <-> Ethernet6

Port Port

From Port	To Port
Modem4	Ethernet6

13. Konfiguriramo WiFi ruter - Internet dobije IP preko DHCP od ISP.

Internet Settings	
IP Configuration	
☒ DHCP	
☐ Static	
☐ PPPoE	
UserName	
Password	
Default Gateway	177.11.10.1
IP Address	177.11.10.2
Subnet Mask	255.255.255.0
DNS Server	8.8.8.8

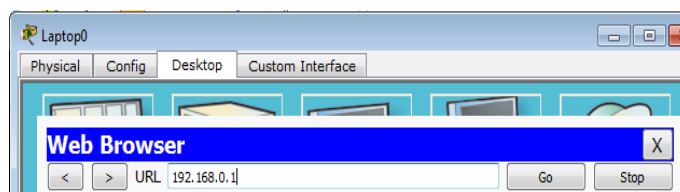
adresu

14. Laptop nema mrežnu karticu. Treba ga ugasiti, izvaditi Ethernet NIC i ubaciti bežičnu sa antenom. Upaliti

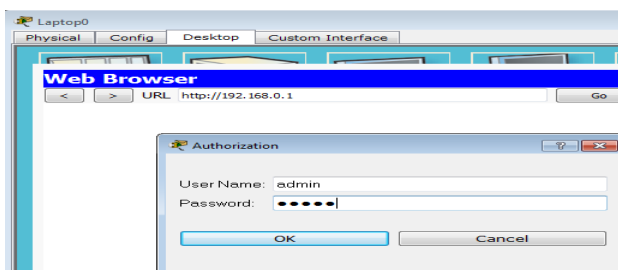


Podesiti da dobije IP adresu od WiFi rutera (obično 192.168.0.101). Probati PING ka bežičnom ruteru i ISP ruteru. Treba da budu uspešni!

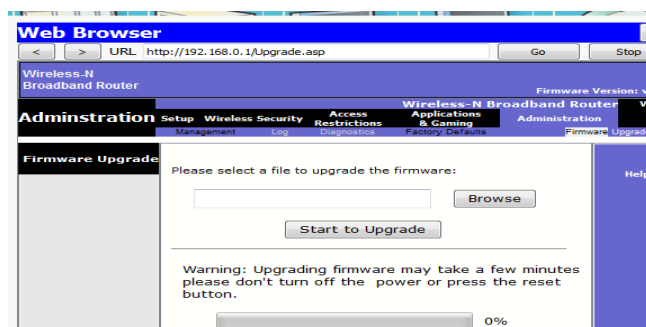
15. Ažuriranje (upgrade) firmvera na bežičnom ruteru radimo sa laptop računara kroz brauzer:



Prijaviti se : admin/admin

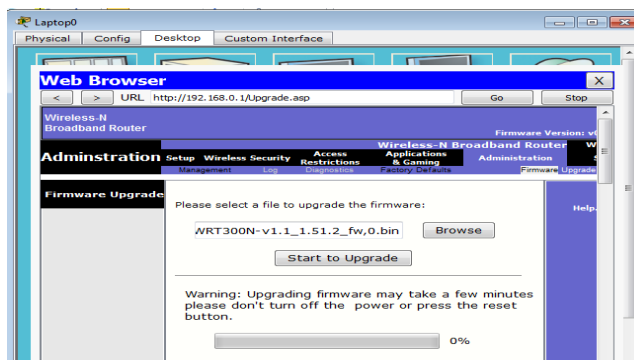


Otići na **Adminstracija -Firmware Upgrade:**

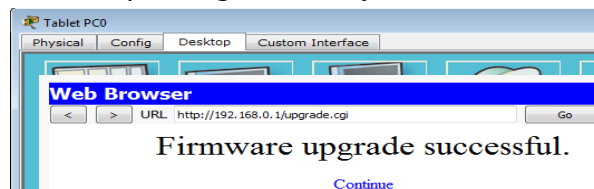


Pronaći odgovarajući fajl (*Browse*)

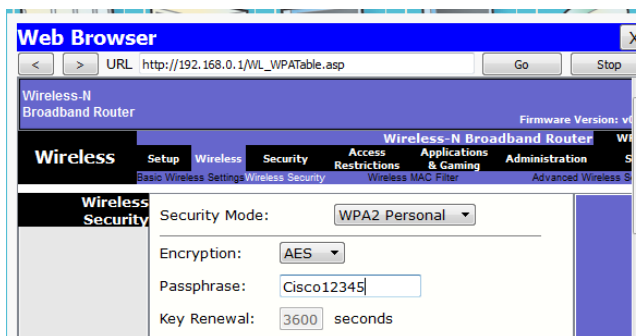
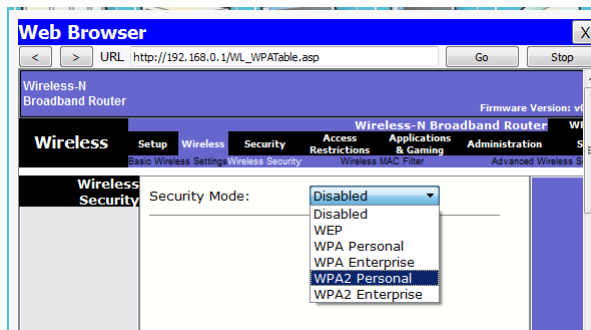
Odabrati fajl (npr. najnoviji) i kliknuti **Start to Upgrade**



Nakon uspešnog ažuriranja firmvera bežičnog rutera dobiaj se:

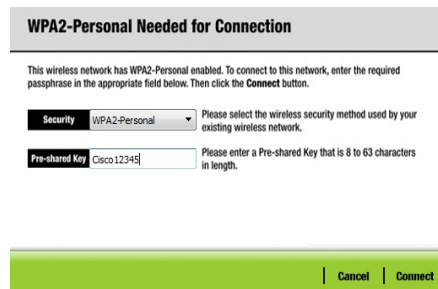
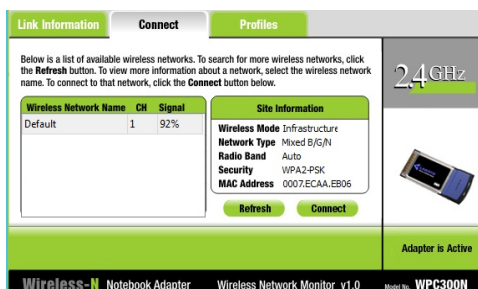


16. Podesiti WIFI ruter za WPA2-Personal. U Meniju - **Wireless - Wireless Security**. Podesiti lozinku - Cisco12345



Pri dnu strane kliknuti na - *Save settings*.

17. Na Laptopu treba podesiti lozinku za pristup SSID. Kliknuti na **PC wireless**, potom na tab Connect i tipku Connect.



Uneti lozinku - *Pre-shared Key* (Cisco12345) i kliknuti na *Connect*

18. Za daljinski pristup potrebno je konfigurisati (odnosno proveriti da li je već konfigurisan pristup preko TELNET-a) za svičeve i rutere:

R1, R2, R3 : Vidimo da nije konfigurisano <i>line con 0</i> <i>line vty 0 4</i> <i>login</i>	Treba: Za Telnet: <i>line vty 0 4</i> <i>password telnet</i> <i>login</i> Za konzolu <i>line con 0</i> <i>password console</i> <i>login</i>	Treba: <i>enable secret admin</i> <i>enable password cisco</i>
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19. Sačuvati konfiguracije svih rutera i svičeva (#**write**)

20. Sačuvati konfiguracije na traženi direktorijum. Popuniti obrasce Прилог док2.doc , Прилог док3.doc i Прилог док4.doc